

Ensemble Learning & Model Optimization Report

Breast Cancer Classification Pipeline

Maincrafts Internship Task 5

1 Objective

This task moves from single classifiers to ensemble methods, building a complete ML pipeline on the Breast Cancer dataset. Logistic Regression is used as a baseline, compared against Random Forest and Gradient Boosting, followed by hyperparameter tuning via GridSearchCV and a final justified model selection.

2 Methodology

- **Dataset:** scikit-learn's Breast Cancer dataset (569 samples, 30 features), target: malignant vs. benign.
- **Split:** 80/20 train-test, `random_state=42`, stratified to preserve class balance.
- **Preprocessing:** `StandardScaler` applied for the Logistic Regression baseline (tree-based models are scale-invariant, so raw features were used for Random Forest and Gradient Boosting).
- **Models trained:**
 - Logistic Regression – linear baseline
 - Random Forest – bagging ensemble, reduces overfitting via averaging many trees
 - Gradient Boosting – sequential ensemble, corrects errors of prior weak learners
- **Hyperparameter tuning:** `GridSearchCV` on Random Forest over `n_estimators` $\in \{50, 100\}$ and `max_depth` $\in \{3, 5, 7\}$, 3-fold CV.

3 Model Comparison

Model	Accuracy
Logistic Regression	0.9825
Random Forest	0.9561
Gradient Boosting	0.9561
Tuned Random Forest	0.9561

Table 1: Accuracy comparison across baseline, ensemble, and tuned models.

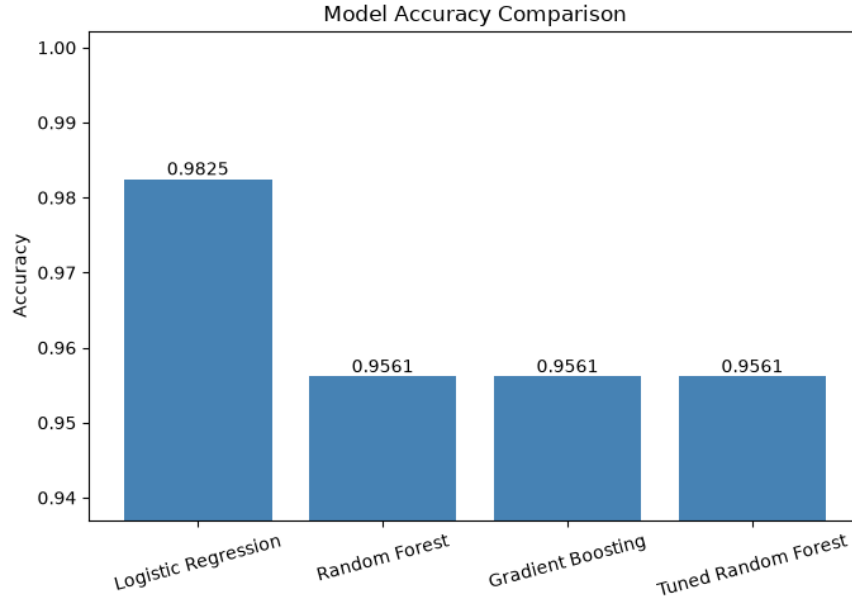


Figure 1: Accuracy comparison across all models.

4 Optimization Explanation

GridSearchCV was used to systematically search combinations of `n_estimators` and `max_depth` for the Random Forest model, using 3-fold cross-validation to avoid overfitting to a single split. The best configuration found was:

Hyperparameter	Best Value
<code>n_estimators</code>	100
<code>max_depth</code>	5

Table 2: Best hyperparameters selected by GridSearchCV.

Tuning `max_depth` controls the bias-variance trade-off within each tree: shallower trees reduce overfitting risk but may underfit, while `n_estimators` improves stability by averaging over more trees at the cost of training time. Despite tuning, the Random Forest accuracy did not exceed its untuned counterpart on this test set.

5 Final Decision

Based on accuracy, stability, and interpretability, the **baseline Logistic Regression** model is selected as the final model:

- **Performance:** achieved the highest accuracy (0.9825) of all four models, outperforming both ensemble methods and the tuned Random Forest (all 0.9561).
- **Stability:** on this dataset, the linear decision boundary generalizes at least as well as the ensemble models, indicating the underlying relationship between features and diagnosis is close to linearly separable after scaling.

- **Interpretability:** Logistic Regression coefficients are directly interpretable, offering a clear advantage over the ensemble models without any accuracy trade-off in this case.

This result illustrates an important real-world lesson: more complex models (ensembles) do not automatically outperform simpler baselines, especially on smaller, well-behaved, linearly-separable datasets like Breast Cancer. The added complexity of Random Forest and Gradient Boosting was not justified here.

6 Conclusion

This task demonstrated a complete ensemble learning and optimization pipeline, but the outcome was a reminder rather than a confirmation: ensemble methods did not outperform a well-scaled Logistic Regression baseline on this dataset. Systematic comparison and hyperparameter tuning still added value – not by improving accuracy, but by providing evidence-based confidence that the simpler model was in fact the correct choice.